On Solving the Multiple Variable Gapped Longest Common Subsequence Problem

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Abstract. This paper addresses the Variable Gapped Longest Common Subsequence (VGLCS) problem, a generalization of the classical LCS problem involving flexible gap constraints between consecutive solutions' characters. The problem arises in molecular sequence comparison, where structural distance constraints between residues must be respected, and time-series analysis, in which events are required to occur within specified temporal delays. We propose a novel search framework based on a root-based state graph representation, in which the overall state space comprises of generally large number of root-based state subgraphs. To cope with the resulting combinatorial explosion, an iterative beam search strategy is employed, dynamically maintaining a global pool of promising candidate root nodes, enabling effective control of diversification across iterations. To explore the search, several heuristic evaluations inspired by the LCS literature are incorporated into the standalone beam search procedure. To the best of our knowledge, this work presents the first comprehensive computational study of the VGLCS problem, generating a benchmark of 320 synthetic instances with up to 10 input sequences and 500 characters. Experimental results show robustness of the novel approach over the baseline beam search in a comparable runtime.

Keywords: Longest common subsequences · Beam Search · Bioinformatics · Gap constraints

1 Introduction

The *Longest Common Subsequence Problem* (LCSP) [1, 4] is a well-known combinatorial optimization problem with numerous applications in bioinformatics

and computational biology, playing a fundamental role in the analysis and comparison of molecular sequences. Given an arbitrarily large set of input sequences $S = \{s_1, \ldots, s_m\}, m \in \mathbb{N}$ over an alphabet Σ, the objective is to identify a longest possible subsequence that is common to all sequences $s_i \in S$. Over the past decades, a variety of practically motivated extensions of the LCSP have been proposed to better capture structural, biological, or application-specific requirements of real-world sequences. Notable variants include constrained, arc-preserving, and repetition-free longest common subsequence problems [7, 6, 3], among others. In this work, we focus on yet another practical variant, the *Variable Gapped LCS Problem* (VGLCSP), originally introduced for the $m = 2$ case in [10], later investigated from a theoretical perspective in [2]. The VGLCSP extends the classical LCSP by incorporating flexible distance constraints (referred to as *gaps*) between consecutive symbols of the resulting subsequence. Unlike fixed-gap models, these gap limits are allowed to vary along the input sequences, thereby offering increased modeling flexibility.

Formally, for each input sequence $s_i \in S$, a gap constraint is defined by a function at positions of sequence as $G_i \colon \{1, \ldots, |s_i|\} \mapsto \mathbb{N}$. Precisely, if the characters of solution s occur at positions $i_i^1 < \cdots < i_i^{|s|}$ in s_i, the gap constraint is fulfilled iff $i_i^x - i_i^{x-1} \leq G_i[i_i^x] + 1, \quad x = 2, \ldots, |s|$. A common subsequence s is considered *feasible* if the corresponding gap constraints are satisfied simultaneously for all sequences in S. As an example of a VGLCSP, given is a set of two input sequences $S = \{s_1 = \texttt{ABCA}, s_2 = \texttt{ACAB}\}$ with two gap constraints $G_1(\{1, \ldots, 4\}) \mapsto \mathbf{1} = G_2(\{1, \ldots, 4\})$. The sequence $s = \texttt{ACA}$ is a feasible solution because it is a common subsequence of the both input strings, where its corresponding characters appear at positions 1, 3 and 4 in s_1 and at the positions 1, 2, and 3 in the s_2. As the difference between each consecutive pair of the indices of these positions is less or equal $1 + 1 = 2$, s represents a feasible sequence of the tackled problem. On the other hand, sequence $s =$AB is also a common subsequence, but not fulfilling all gap constraints. In details, the corresponding characters appear at positions 1 and 2 in s_1 and at positions 1 and 4 in s_2. Obviously, $4 - 1 = 3 > 2 = 1 + 1$ thus violating the second gap constraint.

The VGLCSP provides a flexible and realistic model for sequence comparison, particularly suited to applications in DNA and protein analysis where variable structural distances between (active) residues must be respected. Beyond bioinformatics, the problem also arises in time-series analysis, especially in settings where events are required to occur within specified temporal delays [8].

Although several dynamic programming approaches exist for the two-sequence case ($m = 2$) scenario [10], their computational complexity becomes prohibitive when expanded for larger m. To the best of our knowledge, no effective exact or approximate approaches are currently existing for the generalized VGLCSP with arbitrary large set S, which is clearly an $\mathcal{NP}$-hard problem as a general version of the LCS problem with arbitrary large set of input strings.

Preliminaries and Notation. Let $|s|$ denote the length of a sequence s. The symbol $s[i]$ refers to the character at position i of sequence s, where indexing starts at $i = 1$. The substring of s that begins at position i and ends at po-

sition j is denoted by $s[i,j]$. If $i = j$, this substring corresponds to the single character $s[i]$; if $j < i$, then $s[i,j] = \varepsilon$, where ε denotes the empty string. The number of occurrences of $a \in \Sigma$ in a sequence s is denoted by $|s|_a$. We denote by $S = \{s_1, \ldots, s_m\}$ the set of input sequences, and by $S^{rev} = \{s_1^{rev}, \ldots, s_m^{rev}\}$. Unless stated otherwise, $m \in \mathbb{N}$ represents the number of input sequences (and, correspondingly, the number of gap constraints), and n denotes the length of the longest sequence in S. Given is a positional vector $\mathbf{p}^L \in \mathbb{N}^m$, subproblem related to these positions is given by $S[\mathbf{p}^L] = \{s_i[\mathbf{p}_i^L] \mid i = 1, \ldots, m\}$. Last but not least, for a gap constraint G_i, its reverse constraint is denoted by G_i^{rev}, and is given as $G_i^{rev}(j) := G_i(|s_i| - j + 1)$, for each $j = 1, \ldots, |s_i|$.

The remainder of the paper is organized as follows. Section 2 introduces the rooted state graph representation of the problem. In Section 3, we design of our main methodological contribution, the iterative multi-source beam search approach. Section 4 reports the computational results, comparing the proposed method with a baseline beam search and an iterative greedy heuristicFinally, Section 5 concludes the paper and outlines directions for future research.

2 Rooted State Space Formulation

Building on the state-space representation for the LCSP introduced in [1], we develop a *rooted state graph* model for the VGLCSP. Each state represents one or more feasible partial solutions characterized by a vector of positions induced by the input sequences and by the current subsequence length.

Formally, a partial subsequence s^v induces a state node $v = (\mathbf{p}^{L,v}, l^v)$, where $\mathbf{p}^{L,v} = (p_1^{L,v}, \ldots, p_m^{L,v})$ and the following conditions hold: (i) for each sequence $s_i \in S$, the index $p_i^{L,v} - 1$ is the smallest position such that s^v is a subsequence of $s_i[1, p_i^{L,v} - 1]$; (ii) $l^v = |s^v|$ denotes the length of the partial subsequence; and (iii) all gap constraints G_i are *satisfied* by s^v with respect to each input sequence s_i. A direct arc $\alpha = (v_1, v_2)$, labeled with $\texttt{let}(\alpha) = a \in \Sigma$, exists if $l^{v_2} = l^{v_1} + 1$ and $s^{v_2} = s^{v_1} \cdot a$, that is, state v_2 corresponds to extending a partial solution of v_1 by appending the character a, respecting the gap constraints.

To expand a state v, those characters $a \in \Sigma$ that occur in the suffix of each sequence s_i starting at position $p_i^{L,v}$, for $i = 1, \ldots, m$, are first detected. For each such $a \in \Sigma$, we identify the smallest feasible positions $p_i^{L,a} \geq p_i^{L,v}$ such that $s_i[p_i^{L,a}] = a$ and $p_i^{L,a} - p_i^{L,v} \leq G_i(p_i^{L,a}), i = 1, \ldots, m$. In that way, a child state $w = (\mathbf{p}^{L,a} + \mathbf{1}, l^v + 1)$ is created, unless it is dominated by another child state according to the dominance criteria defined later in this section. To efficiently compute children and their associated position vectors, we employ a preprocessing data structure denoted by `Succ`: a three-dimensional integer array indexed by (i, j, a), where i refers to the input sequence s_i, $1 \leq j \leq |s_i|$ denotes a position within that sequence, and $a \in \Sigma$ is a character. As already pointed out, the value $q = \texttt{Succ}[i, j, a]$ stores the smallest index $q \geq j$ such that $s_i[q] = a$ so that the gap constraint is satisfied at position q, that is, $q - j \leq G_i(q)$. If no such position exists, the value -1 is assigned. Goal states are those that

cannot be further expanded. The trivial root state is given by $r = ((1, \ldots, 1), 0)$ as in-going edges to this node are not possible, representing an empty partial solution as in the case of LCS problem. However, it is not the only root node, in general. The problem can have many (possibly exponentially) root nodes due to the definition of the problem itself allowing a freedom to choose leading letter of partial solutions, induced from the position $u = (i_1^1, \ldots, i_m^1)$ matching that letter, as gap constraints do not apply to the starting letter of each solution.

This motivate us to introduce the state space subgraph rooted with the match $p^{L,u}$, denoted by $\textsc{RootedSubgraph}(u = (p^{L,v}, 0))$, induced by empty partial solution, which consists of all nodes (and corresponding transitions) that can be feasibly reached via node $(p^{L,v} + \mathbf{1}, 1)$, constructed directly from the root node u through its extension by the letter at the position $p^{L,u}$. In terms of feasible partial solutions, $\textsc{RootedSubgraph}(u)$ will generate solutions whose leading letter is the letter at the match $p^{L,u}$, recursively repeating the procedure of expansion of solutions by one letter in all possible ways, until no more expansions are possible. As discussed in the following section, the structure of this subspace heavily depends on the choice of the root state from which the search is initiated. As an example, we illustrate the state space of $\textsc{RootedSubgraph}(u = (1, 1), 0)$, depicted in Fig. 1. Note that here the position $(1, 1)$ correspond to the match of letter `A`, where $u = ((1, 1), 0)$ indeed represents a root node.

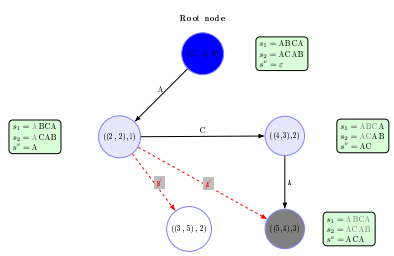


Fig. 1: State space $\textsc{RootedSubspace}(u = ((1, 1), 0))$ between the sequences $\boxed{\texttt{A}}$`BCA` and $\boxed{\texttt{A}}$`CAB`, assuming $G_1 = G_2 = \mathbf{1}$. A final/terminal state is displayed in gray. The white-filled node represents a valid LCS node, but not a feasible VGLCS node, as the second gap constraint has been violated. The longest solution generated from this subspace is `ACA`.

Multi-Source Beam Search. Unlike the classical LCSP, the VGLCSP may exhibit multiple, potentially exponentially many, *disconnected root components* in its complete state space. Consequently, a search strategy that expands states from a single initial root may fail to reach the optimal solution. As an illustrative example given in Fig. 2, consider the sequence set $S = \{s_1 = \texttt{ATGGAAA},\ s_2 = \texttt{ATCCAAA}\}$, with $G_{s_1} = G_{s_2} = 1$.

In this instance, any state with position vector $\mathbf{p}^L = (5, 5)$ cannot be reached from the initial state $((1, 1), 0)$ by a sequence of feasible direct transitions. As

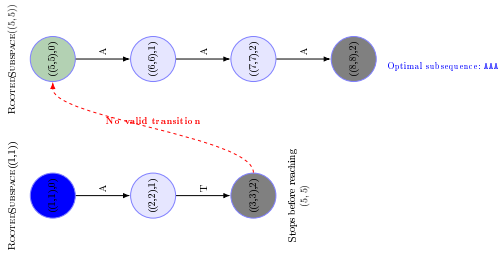


Fig. 2: Root state space subgraphs: disconnected components.

a result, the optimal common subsequence `AAA` is unreachable from the root $((1,1),0)$ when exploring its subgraph. To address this issue, the complete state space of an VGLCSP instance is defined by $\bigcup_{v \in \mathcal{R}} \textsc{RootedSubspace}((\mathbf{q}^{L,v}, 0))$ where $\mathcal{R}$ denotes the set of all root states, which cannot be reached from any other state by a series of direct transitions. However, explicitly enumerating all such root states is computationally prohibitive in general, requiring $O(n^m)$ time, and thus infeasible for instances with large m or n.

This structural characteristic of the VGLCSP motivates the methodology entitled *Iterative Multi-Source Beam Search* (IMSBS) approach, which is specifically designed to dynamically explore multiple promising rooted subregions of the state space by systematically moving from one rooted subgraph to another one. By iteratively identifying and expanding a set of candidate root states, IMSBS will effectively mitigate the disconnection effects induced by gap constraints, enabling the discovery of high-quality solutions that would otherwise remain inaccessible to single-source search methods.

3 Iterative Multi-source Beam Search Framework

Beam search (BS) is a well-established tree-search-based metaheuristic that operates in a breadth-first-search manner. The search is controlled by a beam width parameter β, which specifies the maximum number of nodes retained for expansion at each levels, and by a heuristic guidance h, which guides the selection of most promising candidates for further expansion. Initially, the procedure receives a set of nodes as the initial beam, which are all expanded, generating a set of candidate nodes for a beam of the subsequence level. After being evaluated, up to β most promising nodes are kept for in the beam of the subsequent level. The procedure repeats until beam is empty, returning the longest partial solution.

In the IMSBS, beam search component is served to intensify the search process. As the search space of the VGLCSP is represented by the set of root nodes $\mathcal{R}$, BS procedure is in charge for efficiently exploring the search subspace $\textsc{RootedSubspace}((\mathbf{p}^L, 0))$ induced by the corresponding root node. Note that one could here employ an exact technique (such as A* or BFS); oowever, solving a subproblem itself could be a challenge task, as hard as the original problem itself. Thus, we have opted for more robust BS technique.

The overall IMSBS framework is summarized as follows. Initially, the global pool of candidate root nodes is initialized to an empty set $\mathcal{R} = \{\}$ and the current best solution to $s_{\text{best}} = \varepsilon$. First, for each $a \in \Sigma$ the closest matches $\mathbf{p}^{L,a}$ from the leading positions of each input sequences are detected. Each such (non-dominated) match generates one root node $(\mathbf{p}^{L,a}, 0)$, which has been subsequently added to $\mathcal{R}$. The core iterations of the IMSBS are then executed until either $\mathcal{R}$ becomes empty, a predefined maximum number of iterations is reached, or allowed time limit is exceeded. At each IMSBS iteration, a fixed number (*sources_from_R*) of most promising candidate root nodes according to a heuristic criterion h' are taken from $\mathcal{R}$, subsequently stored in a temporary beam $\mathcal{L}$. These nodes are further refined, executing a single backward beam search procedure for each $w \in \mathcal{L}$ with a root node $(\mathbf{p}^w, 0)$. The procedure works similarly as the baseline BS, expanding nodes with accordance to the gap constraints but in the reversed manner, starting from the root positions at each sequence and moving the search (pointers) towards their leading positions. In essence, for each $w \in \mathcal{L}$, the same RootedBeamSearch is executed on the reversed input sequences S^{rev} and the reversed gap constraints $(G_i^{rev})_{i=1}^m$ with the initial beam width $B = \{((|s_i| - p^w)_{i=1}^m, 0)\}$. Assume the procedure has completed, returning the solution s_{bwrd}^w. In this way, the reverse sequence s_{bwrd}^{rev} is a feasible VGLCS solution. The partial solution without the last letter—which correspond to the match $\mathbf{p}^w$—is assigned to the refined node $w \in \mathcal{L}$, while the updated $w = (p^w, |(s_{bwrd}^w)^{rev}| - 1)$ is passed to the forward beam search procedure to another round of extensions. Thus, the refined set $\mathcal{L}$ enters in as the initial beam, executing a beam search with a beam width β and a heuristic $h(v)$. During this phase, all completed (i.e., non-expandable) nodes are collected in a list $V_{complete}$, being returned along with the found solution s_b after its completion, checked for a new incumbent. Further, if a complete node from $V_{complete}$ corresponds to a solution that improves upon the current best, the solution becomes a new incumbent. Each such node is then independently expanded—ignoring the gap constraints— generating a set of candidate root nodes. In details, for each complete node $v = (\mathbf{p}^v, l^v)$, candidate root nodes $r^{w,a} = (\mathbf{p}^{w,a}, 0)$ are generated for each letter $a \in \Sigma$ consisting of positions $p_i^{w,a}$ that matches letter a in s_i which are the closest (but not equal) to the position $\mathbf{p}_i^v$. Each such candidate node $(r^{w,a}, 0)$ is added in the pool $\mathcal{R}$, provided that it has not already been part during the previous iterations. Once the beam search procedure is terminated, a new Imsbs iteration is initiated with a newly selected set of most promising root nodes from $\mathcal{R}$, until one of the termination condition has met. The pool $\mathcal{R}$ may be maintained as a priority queue implemented via a binary heap, to ensure efficient extraction of the most promising candidate root nodes.

The core advantages of the IMSBS approach over the baseline BS are: (i) A provision of balance between complete beam search execution (local exploration of promising subgraphs) and the generation of new candidate root nodes (global coverage of the search space); (ii) Prevention from suboptimal candidate root nodes reduced by executing backward beam search procedure.

Heuristic guidances. Several heuristic guidances to guide search are utilized

within the IMSBS—as the candidates to guide froward and backward BS, h and h' respectively. These are known from the LCS literature: (i) *"Look-ahead" for the remaining length:* $\mathrm{UB}_1(\mathbf{v}) = l^v + \min_{i=1,\ldots,m}\left(|s_i| - p_i^{L,v} + 1\right)$; (ii) *Character Frequency Alignment:* $\mathrm{UB}_2(\mathbf{v}) = l^v + \sum_{\sigma\in\Sigma} \min\left(|s_1[p_1^{L,v}, |s_1|]|_\sigma, \ldots, |s_m[p_m^{L,v}, |s_m|]|_\sigma\right)$, counting the maximum number of characters that can still match in the (optimal) solution based on the frequency of each character; (iii) *A probability-based heuristic* guidance h_{prob} by the Mousavi and Tabataba's matrices [9]: These probabilities are based on the idea of determining a probability of the event that a sequence s of length k is a subsequence of a (random) sequence of length n over Σ. More about these heuristics the interested readers can found at [1, 9].

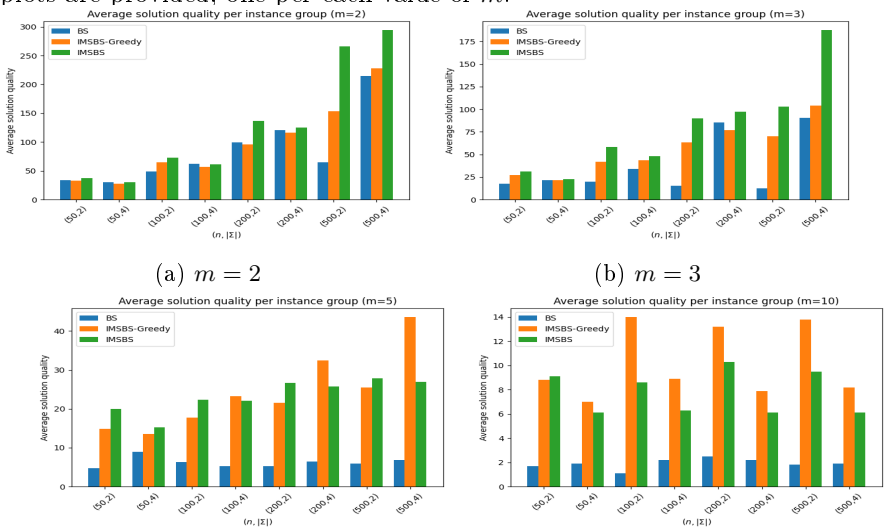
4 Experimental Evaluation

In this section, we compare the performance of heuristic approaches designed for general VGLCSP with an arbitrary $m \geq 2$. Specifically, the following methods are included: (i) BS: a baseline beam search approach that starts from root node r as the initial beam, performing a single iteration of IMSBS while employing a huge beam width; (ii) IMSBS-GREEDY: a variant with a fixed parameter $\beta = 1$ employing a huge number of IMSBS iterations; (iii) IMSBS: a tuned IMSBS configured to employ an average runtime comparable to that of BS.

The IMSBS is implemented in Python 3.11 and evaluated on the VEGA high-performance computing system hosted at IZUM, Maribor. The cluster consists of 960 compute nodes equipped with AMD EPYC 7H12 CPUs operating at 2.35 GHz. All experiments were conducted in a single-threaded setting. For each combination of $n \in \{50, 100, 200, 500\}$, $m \in \{2, 3, 5, 10\}$, and $|\Sigma| \in \{2, 4\}$, 10 random problem instances are generated. First, m sequences of equal length are generated uniformly at random. Then, the gap constraints for each instance are generated such that each value $G_i(j), j = 1, \ldots, |s_i|, i = 1, \ldots, m$ is assigned uniformly from $\mathcal{U}(\{\lfloor 0.5 \cdot |\Sigma| \rfloor, \ldots, \lfloor 1.5 \cdot |\Sigma| \rfloor\})$. Therefore, a total of *320 problem instances* is generated. The benchmark set is denoted as RANDOM, freely available at `https://github.com/markodjukanovic90/NikolaBalabanDiplomski/tree/main/src/instance_i_generator`.

Parameter tuning. To tune the parameters of our algorithms, we employed a basic grid search to more sensible parameters: beam width in the beam search forward pass (β), and heuristic guidance $h \in \{\mathrm{UB}_1, \mathrm{UB}_2, h_{prob}\}$, comparing the average solution quality over all (320) problem instances. Less sensible parameters are chosen after several preliminary runs. We fix $sources_from_R = 10$ and $beam_iters = 100$, and $h' = \mathrm{UB}_2$. For the baseline BS approach, a high beam width $\beta = 10,000$ and $h = h_{\mathrm{prob}}$ are utilized. Remaining parameters of the IMSBS approach received the following values: $h = \mathrm{UB}_2$ and $\beta = 500$ to keep comparable average runtime between IMSBS and the baseline BS approach. These values balance runtime relative to BS while maintaining reasonable solution quality. Similarly, for the IMSBS-GREEDY approach with fixed $\beta = 1$ uses $beam_iters = 10000$, and all other parameters hold as in the tuned IMSBS.

Numerical results for all three heuristic approaches are displayed in Fig. 3, reporting the average solution quality per instance group for each approach. Four plots are provided, one per each value of m.



(a) $m = 2$ (b) $m = 3$

(c) $m = 5$ (d) $m = 10$

Fig. 3: Average solution quality of the three approaches.

The IMSBS approach produces the highest average solution quality in 21 out of 32 instance groups, surpassing the baseline BS approach dramatically, indicating the selection of a right root node is essential in achieving high-quality solutions. The baseline BS performs equally well or better in just one case. Concerning the IMSBS-GREEDY, it surpasses the other two approaches in 10 (out of 32) cases, mostly for the instances with larger $m = 10$, having shorter final solutions where it turns out that frequently moving from one to another root and exploring respective subgraphs of the search space is strategically crucial than just strengthening intensification around a solution with a high beam width. Exploring additional root nodes serves to improve the diversity of the search process, which appears necessary in the case when expected solutions are short. This is exactly where IMSBS-GREEDY fits well. The BS and IMSBS approaches are tuned to exhibit comparable average runtimes. IMSBS-GREEDY has a significantly larger average runtime due to a higher number of executed iterations ($beam_iters = 10,000$) for the instances with larger m, in contrast to by iterations less intensive approaches.

5 Conclusions and Future Work

In this work, we studied a generalized version of the Longest Common Subsequence (LCS) problem with an arbitrary number of sequences, called the variable

gapped LCS problem. It extends the classical LCS by constraining the allowable distances between matched characters, reflecting biological scenarios in which spatially closer nucleotides interact more strongly than distant ones We proposed an enhanced Iterative Multi-Source Beam Search (IMSBS) approach which performs local exploration from promising root nodes involving a global strategy of selecting candidate root nodes based on previously collected search information. Experimental results indicate the IMSBS approach consistently produces highest-quality solutions over the baseline beam search strategy. In the case of instances with shorter feasible sequences, the strategy of frequently moving from one feasible subspace to another one appears more efficient, than in the case of small number of sequences, where a beam search of reasonably high beam width and IMSBS iterations is necessary in achieving high-quality solutions.

Future research includes designing more sophisticated heuristics that directly incorporate gap constraints into scoring, optimizing IMSBS to efficiently reuse previously explored regions of the state space, testing designed approaches on real-world biological instances, and scaling experiments up to longer sequences.

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References

1. Finding longest common subsequences: New anytime A* search results. Applied Soft Computing **95**, 106499 (2020)
2. Adamson, D., Kosche, M., Koß, T., Manea, F., Siemer, S.: Longest common subsequence with gap constraints. In: International Conference on Combinatorics on Words. pp. 60–76. Springer (2023)
3. Adi, S.S., Braga, M.D., Fernandes, C.G., Ferreira, C.E., Martinez, F.V., Sagot, M.F., Stefanes, M.A., Tjandraatmadja, C., Wakabayashi, Y.: Repetition-free longest common subsequence. Discrete Applied Mathematics **158**(12), 1315–1324 (2010)
4. Bergroth, L., Hakonen, H., Raita, T.: A survey of longest common subsequence algorithms. In: Proceedings Seventh International Symposium on String Processing and Information Retrieval. SPIRE 2000. pp. 39–48. IEEE (2000)
5. Blum, C., Festa, P.: Metaheuristics for String Problems in Bio-informatics, vol. 6. John Wiley & Sons (2016)
6. Farhana, E., Rahman, M.S.: Constrained sequence analysis algorithms in computational biology. Information Sciences **295**, 247–257 (2015)
7. Jiang, T., Lin, G., Ma, B., Zhang, K.: The longest common subsequence problem for arc-annotated sequences. Journal of Discrete Algorithms **2**(2), 257–270 (2004)
8. Lainscsek, C., Sejnowski, T.J.: Delay differential analysis of time series. Neural computation **27**(3), 594–614 (2015)

9. Mousavi, S.R., Tabataba, F.: An improved algorithm for the longest common subsequence problem. Computers & Operations Research **39**(3), 512–520 (2012)
10. Penga, Y.H., Yangb, C.B.: The longest common subsequence problem with variable gapped constraints. In: Proceedings of the 28th Workshop on Combinatorial Mathematics and Computation Theory, Penghu, Taiwan. pp. 17–23 (2011)

A ILP model for the fixed $m = 2$ VGLCS Problem

When considering the VGLCS problem with two input sequences, in this section we propose an *integer programming model*, representing a methodological approach that differs from most existing approaches in the literature, which are predominantly based on dynamic programming (DP). This is a proof-of-concept showing structural difficulty of this approach and a baseline modeling contribution when solving the fixed version of the tackled problem. Shortly, the purpose of the ILP model is not computational competitiveness but structural modeling and benchmarking.

To this end, let:

- $M = \{(i, j) \mid s_1[i] = s_2[j]\}$ denote the set of all positions where the characters match.

For each $(i, j) \in M$, we define a binary variable:

$$x_{i,j} = \begin{cases} 1, & \text{if the pair } (i,j) \text{ is selected as part of the solution,} \\ 0, & \text{otherwise.} \end{cases}$$

We introduce additional binary variables $s_{i,j}$ indicating whether the pair (i, j) represents the *starting position of the subsequence*.

Objective function. We maximize the number of selected admissible matches:

$$\max \sum_{(i,j)\in M} x_{i,j}$$

Constraints. The following constraints are imposed:

1. Predecessors (variable gap). For each $(i, j) \in M$, we define the set of valid predecessors:

   $$\text{Pred}(i, j) = \{(i', j') \in M \mid i' < i,\ j' < j,\ i - i' \leq G_{s_1}[i] + 1,\ j - j' \leq G_{s_2}[j] + 1\}$$

   The activation constraint is given by:

   $$x_{i,j} \leq \sum_{(i',j')\in \text{Pred}(i,j)} x_{i',j'} + s_{i,j}$$

2. Starting position. At most one starting pair is allowed:

   $$\sum_{(i,j)\in M} s_{i,j} \leq 1$$

3. Conflict constraints (character ordering). Two distinct pairs (i, j) and (i', j') are in conflict if they violate the character order:

   $$\text{If } (i \leq i' \text{ and } j \geq j') \text{ or } (i \geq i' \text{ and } j \leq j'), \text{ then: } \quad x_{i,j} + x_{i',j'} \leq 1$$

4. Auxiliary constraints: $s_{i,j} = 1 \Rightarrow x_{i',j'} = 0, \forall (i' \leq i-1, j' \leq j-1)$. The constraints are expressed in the following form:

$$\sum_{(i',j')|i' \leq i \wedge j' \leq j} x_{i'j'} + s_{i,j} \leq 1, \forall (i,j) \in M$$

The domain of the variables is given by

$$x_{i,j},\ s_{i,j} \in \{0,1\}, \quad \forall (i,j) \in M.$$

The model is formulated to maximize the number of valid matches forming a common subsequence, while respecting the ordering of elements and the maximum allowed gaps between consecutive elements in both sequences.

B Pseudocodes

Algorithm 1 Rooted Beam Search

```
function RootedBeamSearch(R, S, {G_i}_{i=1}^m, β, h)        ▷ r: root node,
S = {s_1, ..., s_m}, gap constraints G_i, beam width β, heuristic h
    s_best ← ε
    B ← R                                                   ▷ Starting beam
    V_complete ← ∅
    while B ≠ ∅ do
        Children ← ∅
        for all v ∈ B do
            if v is complete then
                V_complete ← V_complete ∪ {v}
                if v.l^v > |s_best| then
                    s^v ← Extract partial solution associated with v
                    s_best ← s^v
                end if
                continue
            end if
            C_v ← Expand(v, S, {G_i}_{i=1}^m)
            Children ← Children ∪ C_v
        end for
        Sort(Children, h)                                   ▷ decreasing order
        B ← Children[: β]
    end while
    return s_best, V_complete
end function
```

Algorithm 2 Iterated Multi-source Beam Search (IMSBS) Framework

Require: Sequences $s_1, \ldots, s_m$; Gap functions $G_{s_i}(\cdot)$; beam width $\beta > 0$; $sources_from_R > 0$; $beam_iters > 0$, heuristics h, h', time limit t_{lim}
Ensure: Approximate common subsequence s_{best} which fulfills the gap constraints

Initialize $\mathcal{R} \leftarrow \{(\mathbf{1}, 0)\}$ ▷ A set of root nodes
for all $a \in \Sigma$ **do** ▷ Find closest matching positions for each letter
  $\mathbf{p}_a^w = (p_{i,a}^w)_{i=1}^m \leftarrow$ minimum position $p_{i,a}^w \geq 1$ and $s_i[p_{i,a}^w] = a$ (ignore all gap constraints, remove dominated root positions)
  $R \leftarrow R \cup \{(p_a^w, 0)\}$
end for
$iter \leftarrow 0$
$s_{\text{best}} \leftarrow \varepsilon$
while $\mathcal{R} \neq \emptyset \wedge iter < beam_iters \wedge t_{lim}$ not exceeded **do**
  Select $\mathcal{L} \subseteq \mathcal{R}$ $sources_from_R$ and remove them from $\mathcal{R}$ ▷ $\mathcal{R}$ prioritized by h'
  for all $w \in \mathcal{L}$ **do**
    $s_{bwrd}^w, _ \leftarrow$ RootedBeamSearch($(\{(|s_i| - p_i^w - 1)_{i=1}^m, 0\}, S^{rev}, G^{rev}, \beta', h')$ ▷ Refine each $w \in \mathcal{L}$
    $l^w \leftarrow |(s_{bwrd}^w)^{rev}|$ ▷ Update l^w, assign the reverse of s_{bwrd}^w to w
  end for

  $s_b, V_{complete} \leftarrow$ RootedBeamSearch($\mathcal{L}, S, G, \beta, h$)
  if $|s_b| > s_{best}$ **then**
    $s_{best} \leftarrow s_b$
  end if
  for all $v \in V_{complete}$ **do**
    for all $a \in \Sigma$ **do** ▷ Find closest matching positions for each letter
      $\mathbf{p}_a^w = (p_{i,a}^w)_{i=1}^m \leftarrow$ minimum position $p_{i,a}^w > p_i^v$ and $s_i[p_{i,a}^w] = a$ (ignore all gap constraints)
      $\mathcal{R} \leftarrow \mathcal{R} \cup \{r^w = (\mathbf{p}_a^w, 0)\}$ (if has not previously been added) ▷ Update $\mathcal{R}$
    end for
  end for
  $iter \leftarrow iter + 1$
end while
return s_{best}

C Experimental Evaluation: Additional plots

Table 1: Numerical results of the three heuristic approaches on the Random benchmark suite.

Inst.			Bs		Imsbs-Greedy		Imsbs	
m	n	$\lvert\Sigma\rvert$	$\overline{obj}$	$\bar{t}[s]$	$\overline{obj}$	$\bar{t}[s]$	$\overline{obj}$	$\bar{t}[s]$
2	50	2	33.6	0.02	33.1	0.00	**37.7**	0.06
2	50	4	30.1	0.98	27.7	0.00	**30.1**	0.16
2	100	2	48.9	2.07	64.5	0.01	**72.8**	0.94
2	100	4	**62.1**	11.19	56.9	0.01	61.6	0.91
2	200	2	99.1	18.56	95.5	0.02	**136.4**	6.21
2	200	4	120.5	38.15	116.1	0.05	**124.9**	6.58
2	500	2	65.3	23.27	153.6	0.07	**265.7**	119.75
2	500	4	214.6	163.69	227.7	0.12	**294.4**	60.49
3	50	2	17.5	0.03	27.2	0.00	**31.2**	0.14
3	50	4	21.7	0.18	21.5	0.00	**22.9**	0.19
3	100	2	19.7	0.06	41.8	0.01	**58.5**	3.15
3	100	4	34.1	2.35	43.4	0.03	**48.4**	5.58
3	200	2	15.2	0.16	63.6	0.02	**90.0**	22.48
3	200	4	85.3	23.45	77.2	0.08	**97.1**	72.25
3	500	2	12.6	0.10	69.9	0.07	**102.9**	53.56
3	500	4	90.7	86.35	104.2	0.29	**187.7**	412.27

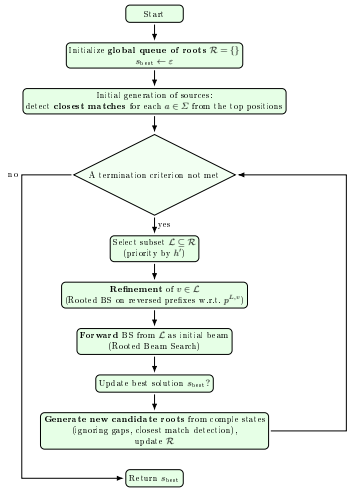
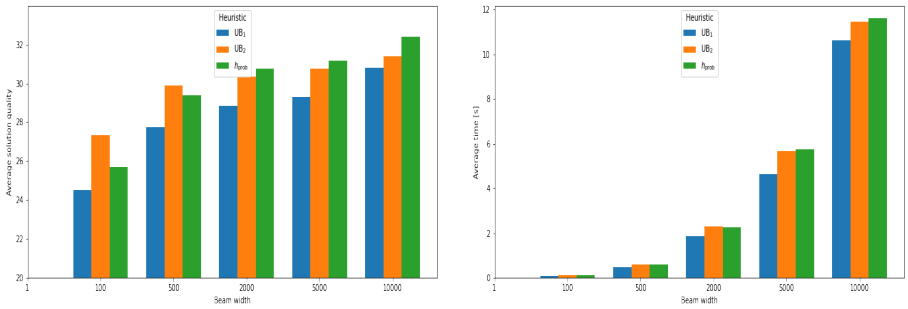


Fig. 4: Simplified workflow of the Iterated Multi-source Beam Search (IMSBS) framework.



(a) Avg. quality over all instances from the RANDOM benchmark suite.

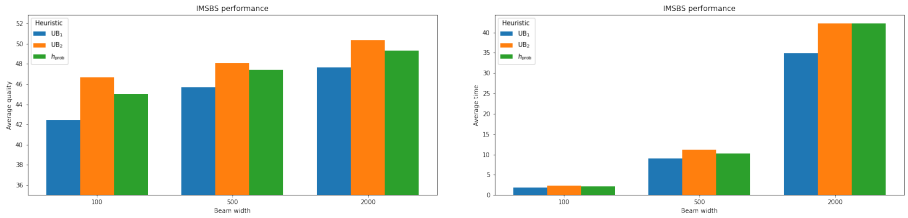
(b) Avg. runtime over all instances from the RANDOM benchmark suite.

Fig. 5: Beam search tuning results on the RANDOM benchmark suite.

D Numerical results for the special case $m = 2$

In this section, we compare the results of approaches from the literature that are specialized for the case $m = 2$. The competitors are listed below and are described in detail in [10].

- DP: the basic dynamic programming algorithm;
- DP-1: an advanced dynamic programming approach that uses Incremental Suffix Maximum Queries (ISMQ) with `Col` and `All` matrices to accelerate the basic DP algorithm;
- DP-2: an enhanced dynamic programming algorithm that handles ISMQ using a *dequeue* data structure for further speedup. All three DP approaches



(a) Avg. quality for different IMSBS settings over all instances from the RANDOM benchmark suite.

(b) Avg. runtime for different IMSBS settings over all instances from the RANDOM benchmark suite.

Fig. 6: IMSBS parameter tuning results on the RANDOM benchmark suite.

are from [10]. Note that the original paper proposes answering ISMQ using Union-Find operations; however, due to lack of implementation details, we decided to employ a dequeue-based approach in our implementation.

– ILP: an integer linear programming method proposed in this work (see Appendix A), motivated by the ILP model for LCS problem [5].

Each algorithm was allocated 30 minutes of execution time. The ILP model was solved using the general-purpose solver CPLEX version 22.1.

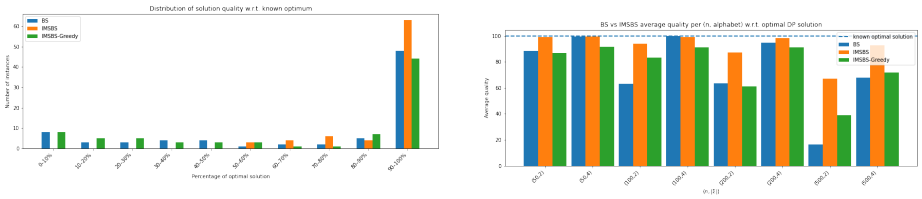
The results are reported in Table 2, which is organized as follows. The first three columns describe the characteristics of the instance groups, over which the results are averaged (10 instances per group). The remaining columns report the performance of the four algorithms: DP, DP-1, DP-2, and ILP. For each approach, both the average solution quality and the average execution time are provided across the 10 instances.

Table 2: Results on the RANDOM benchmark set for $m = 2$: the exact approaches from the literature.

Inst.			DP		DP-1		DP-2		ILP	
m	n	$\lvert\Sigma\rvert$	$\overline{obj}$	$\bar{t}[s]$	$\overline{obj}$	$\bar{t}[s]$	$\overline{obj}$	$\bar{t}[s]$	$\overline{obj}$	$\bar{t}[s]$
2	50	2	38.1	0.01	38.1	**0.01**	38.1	0.01	38.1	168.3
2	50	4	30.3	**0.01**	30.3	0.02	30.3	0.02	30.3	28.0
2	100	2	77.4	0.1	77.4	**0.03**	77.4	0.05	–	–
2	100	4	62.3	0.07	62.3	**0.06**	62.3	0.09	0.00	1800.0
2	200	2	156.4	0.75	156.4	**0.13**	156.4	0.16	–	–
2	200	4	127.2	0.59	127.2	**0.25**	127.2	0.32	–	–
2	500	2	395.9	13.57	395.9	**0.84**	395.9	1.05	–	–
2	500	4	317.2	10.18	317.2	**1.70**	317.2	2.1	–	–

Based on the numerical results presented in Table 2, the following conclusions can be drawn:

- All three dynamic programming approaches are highly efficient, solving all 80 problem instances.
- The runtimes of DPs required to obtain optimal solutions are relatively short, mostly remaining below 10 seconds.
- The ILP approach is able to solve only the smallest instances with $n = 50$; its execution times are roughly two orders of magnitude higher than those of the DP approaches. For larger instances, especially those with $n = 200$, the ILP approach is unable to solve even the initial (root) relaxations.



(a) Relative solution quality percentage distribution for each heuristic approach with respect to the optimal solutions.

(b) Relative solution quality achieved by the heuristic approaches compared to the optimal solutions.

Fig. 7: Number of instances for which BS and IMSBS achieve a specific ratio of the obtained (approximate) solution quality to the known optimal solution for $m = 2$.

Concerning efficiency of BS, IMSBS-GREEDY, and IMSBS on the instances solved optimally by dynamic programming ($m = 2$), Figure 7 illustrates the distribution of instances (on the y-axis) achieving specific ratios of solution quality relative to the known optimal solutions (on the x-axis). In particular, IMSBS reaches a near-optimal solution in 63 out of 80 instances. These numbers are significantly lower in the case of other two heuristic approaches. While in the case of instances with $|\Sigma| = 4$ (and $m = 2$), difference in the solution qualities between these two approaches is slightly in favor of IMSBS, in the case of instance with small-sized alphabet ($|\Sigma| = 2$) the IMSBS approach clearly outperforms the other two approaches. Obviously, for the group of instances with $n = 500$ and $|\Sigma| = 2$ there is a room for further improvement. This is due to the fact that the LCS heuristics perform weakly on the instances with smaller alphabet size and larger sequence length [1] thus one has to look for other alternatives, as discussed in the future work section.